

9 REALMS POSTMORTEM MEMO

BIFROST Options v1.2 — Honest Results After Fixes

804 DTE-matched trades · 2022–2026 · Bootstrap 95% CIs on every cell

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HEADLINE — v1.2 HONEST NUMBERS

After applying all ten red team fixes, the blanket BIFROST Options strategy (buy ATM calls on any ODIN-tier catalyst, T-14 entry → T-1 exit) averages -2.12% mid-fill and -7.89% at realistic 40% spread capture across 804 DTE-matched trades. The 95% CI on the 40%-capture number is $[-13.68\%, -1.70\%]$ — fully below zero. A naive options overlay loses money.

Two segment edges survive every stress test: (1) Phase 1/2 positive readouts: $+45.0\%$ mid, $+33.8\%$ at 40% capture, $n=39$, 59% win, CI $[+13.4\%, +79.0\%]$. (2) PDUFA micro-cap all outcomes: $+34.1\%$ mid, $+24.9\%$ at 40% capture, $n=65$, 49% win, CI $[+6.1\%, +66.9\%]$ — but almost entirely tail-driven (strip top 5 → $+4.3\%$).

Most surprising result: the ODIN filter inverts in options. T1+T2 filter → -6.73% mid ($n=225$). T3+T4 ("inverted") → $+14.32\%$ mid ($n=97$). Options premium pays for uncertainty, not quality. We should not be using ODIN tier as an options selection filter.

1. What v1.2 Shipped

v1.2 is a retroactive recompute of the v1.1 1,828-trade dataset using every methodology fix identified in the v1.0 Red Team memo. No new ORATS pulls, no new trade logic — just honest filters and honest fill models applied to the existing per-trade bid/ask/mid/dte data that was already cached.

Fixes applied

- DTE-matched exits only: kept trades where $\text{entry_dte} - \text{exit_dte} \in [13, 23]$ days AND $\text{exit_dte} \geq 3$ days. This removed the "weekly trap" bug where the T-1 exit landed on a nearer-dated contract. 1,828 → 804 trades (44% survival), close to the red team estimate of 830 DTE-matched.
- IV parser hard cap: rejected rows with $|\text{entry_iv_pct}|$ or $|\text{exit_iv_pct}|$ outside (0.01%, 500%), or $|\text{iv_change_pct}| > 1000\%$. This removed 60 trades with REGN-style $10^{14}\%$ blowups. 1,828 → 1,768 after IV filter alone.
- Four fill-model returns per trade: WORST (cross the spread both ways), REAL_25 (25% spread capture), REAL_40 (40% — the realistic paced-limit assumption), MID (50% — textbook fill). All segments publish MID + REAL_40 side-by-side.
- Bootstrap 95% CI on every segment: $n_{\text{boot}}=2000$, seed=42, percentile method. CI-crosses-zero flag on every cell.

- Publishable threshold $n \geq 30$: anything below 30 is reported for reference but not a live-tradeable edge.
- ODIN inversion test: explicit side-by-side T1+T2 vs T3+T4 evaluation.
- Dropped v5_score filter entirely (red team confirmed leakage with +0.44 to +0.52 outcome-conditional separation across all years).
- Top-k robustness: for tail-sensitive strategies (PDUFA micro, Readout Early) publish avg with top 2 and top 5 trades removed.

What did NOT change

- Underlying ORATS bid/ask/mid snapshots — still the v1.1 3,215-response cache.
- Entry/exit date logic — still T-14 trading days → T-1 trading day.
- Strike selection at entry — still ATM call with delta in [0.20, 0.80] and spread_pct ≤ 55%.
- Universe coverage — still excludes nano caps (<\$50M) and illiquid chains (spread > 55%).

1.1 Nothing beats zero on the broad strategy

Scope	n	MID avg	MID 95% CI	REAL_40 avg
All honest trades	804	-2.12%	[-8.10%, +4.40%]	-7.89%
PDUFA only	322	-0.39%	[-10.89%, +11.19%]	-5.82%
Readout only	482	-3.27%	[-10.19%, +4.40%]	-9.28%

The full honest dataset is negative expected value at realistic execution costs. The readout slice in particular has a 40%-capture CI of [-15.97%, -2.07%], entirely below zero. This means: you cannot generically overlay options on every biotech catalyst in your playbook. You must be selective.

2. PDUFA Segments (DTE-Matched)

All 322 DTE-matched PDUFA trades, broken out by cap tier and outcome. Segments with $n < 30$ are shown in italics/greyed and are not live-tradeable.

Segment	n	MID avg	MID 95% CI	Win %	REAL_40 avg
Micro / approve	32	+36.6%	[-7.2%, +92.0%]	53.1%	+28.1%
Micro / crl	33	+31.6%	[-2.7%, +72.2%]	45.5%	+21.9%
Micro / ALL	65	+34.1%	[+6.1%, +66.9%]	49.2%	+24.9%
Small / approve	38	+1.9%	[-16.4%, +21.2%]	42.1%	-4.8%
Small / crl	30	+13.3%	[-12.6%, +41.5%]	43.3%	+3.1%
Small / ALL	68	+6.9%	[-8.4%, +24.0%]	42.6%	-1.3%
Mid / approve	54	-5.3%	[-20.8%, +12.5%]	38.9%	-10.4%
Mid / crl	17	-18.8%	[-47.3%, +15.7%]	23.5%	-23.9%
Mid / ALL	71	-8.5%	[-23.1%, +7.5%]	35.2%	-13.7%
Large / approve	101	-11.1%	[-33.2%, +14.2%]	27.7%	-13.1%
Large / crl	17	-64.2%	[-78.0%, -49.9%]	5.9%	-65.5%
Large / ALL	118	-18.7%	[-37.2%, +1.5%]	24.6%	-20.7%

PDUFA findings

- PDUFA Micro is the ONLY PDUFA cap tier with a MID CI entirely above zero (+6.1% lower bound, $n=65$ ALL). At REAL_40 the CI lower bound touches zero (-1.6%) — the edge is real but fragile.
- PDUFA Small is coin-flip: +6.9% ALL with CI spanning zero, 42.6% win rate. Not a live edge, not a clear loser either.
- PDUFA Mid and Large are consistently negative. Large/CRL at -64.2% is the single worst sub-segment in the dataset — the stock crashes, so the call premium evaporates AND the crash isn't enough to offset the theta loss because the stock was expensive to begin with.
- Counterintuitive: Micro/CRL (+31.6%, 45.5% win) is positive because small-cap CRLs trigger violent moves that the ATM call can ride directionally in either leg of the move's IV crush.

3. Readout Segments (DTE-Matched)

All 482 DTE-matched readout trades, broken out by phase and outcome. Phases are bucketed from raw stage strings — Phase 1/2 combines "Phase 1/2" and "Phase 2a" where they appear.

Segment	n	MID avg	MID 95% CI	Win %	REAL_40 avg
Phase 1 / positive	54	−5.4%	[−24.3%, +16.1%]	25.9%	−13.1%
Phase 1 / ALL	62	−5.5%	[−21.8%, +12.8%]	29.0%	−13.2%
Phase 1/2 / positive	36	+45.1%	[+12.0%, +80.5%]	58.3%	+33.7%
Phase 1/2 / ALL	39	+45.0%	[+13.4%, +79.0%]	59.0%	+33.8%
Phase 2 / positive	113	+2.1%	[−13.2%, +18.6%]	38.9%	−4.6%
Phase 2 / negative	31	−3.0%	[−25.4%, +20.2%]	38.7%	−10.0%
Phase 2 / ALL	144	+1.0%	[−11.7%, +15.2%]	38.9%	−5.8%
Phase 2b / positive	21	−26.8%	[−47.3%, −5.4%]	23.8%	−31.6%
Phase 2b / ALL	25	−31.7%	[−49.9%, −11.6%]	20.0%	−36.2%
Phase 3 / positive	169	−15.0%	[−25.9%, −3.7%]	33.1%	−19.1%
Phase 3 / negative	43	+4.6%	[−19.9%, +34.5%]	37.2%	−0.3%
Phase 3 / ALL	212	−11.1%	[−21.2%, +0.1%]	34.0%	−15.3%

Readout findings

Phase 1/2 is the single robust edge in this entire backtest. +45.1% MID avg, +33.7% at 40% capture, 58.3% win rate, CI fully above zero at both fill models. It survives top-5 removal (see Section 5 strategy table). It is, however, n=36 — a single bad year of Phase 1/2 biotechs could meaningfully move the mean.

- Phase 2 is coin-flip (+2.1% positive, win 38.9%, CI spans zero). A bigger n doesn't rescue it — the edge just isn't there at the phase level. Separating winners from losers will need pre-readout signal (Gungnir v46 probability, IV cheapness, sponsor track record).
- Phase 2b and Phase 3 are consistently negative. Phase 3 has the largest n in the dataset (212 trades) and its 95% CI upper bound is +0.1% — essentially tagging zero from below. Options on Phase 3 readouts lose money in aggregate.
- Counterintuitive: Phase 3 NEGATIVE readouts are FLAT (+4.6%, win 37.2%) — the pre-readout runup into a failure wasn't large enough to make the ATM call worthless when the stock cratered (ATM calls go to near-zero on a −50% stock move regardless).

4. ODIN Filter Inversion — Confirmed

The v1.0 red team hypothesized that the ODIN tier filter — which was selecting T1/T2 catalysts for the announced strategy — was actually INVERTING options edge. v1.2 tests this directly, side-by-side, on the same 322 DTE-matched PDUFA trades.

ODIN Tier	n	MID avg	MID 95% CI	Win %	REAL_40
T1 (≥ 0.85)	121	-10.17%	[-28.13%, +10.70%]	28.9%	-13.22%
T2 (0.65–0.85)	104	-2.73%	[-15.98%, +11.18%]	42.3%	-8.43%
T3 (0.40–0.65)	13	+16.94%	[-35.95%, +94.14%]	23.1%	+11.88%
T4 (< 0.40)	84	+13.92%	[-6.97%, +37.42%]	39.3%	+5.33%
T1+T2 (standard)	225	-6.73%	[-19.30%, +5.95%]	35.1%	-11.01%
T3+T4 (INVERTED)	97	+14.32%	[-6.29%, +38.31%]	37.1%	+6.21%

Reading the table: T1 (highest conviction approvals) → -10.17% MID, 28.9% win, and the CI is skewed heavily negative (-28.13% lower bound). T4 (lowest conviction) → +13.92% MID, 39.3% win, and CI lower bound is only -6.97%. Monotonic inversion across all four tiers.

Gap: 21 percentage points between T1+T2 and T3+T4 at MID. 17 percentage points at 40% capture. This is decisive.

Why this happens

- T1 catalysts have high prior probability of approval. The equity runs up more in advance (Gungnir/BIFROST research confirms 4.88% median runup on micro/nano). IV ramps earlier. By T-14 the option is already rich, and delta+vega are both crowded.
- T4 catalysts are market-labeled as uncertain. IV is not ramped, term structure is flatter, and the T-1 exit either benefits from a pre-event IV spike (vega) or a directional surprise (delta). The absence of consensus is what pays.
- This matches a well-documented regularity in option markets: premiums compensate for uncertainty, not expected direction.

Recommendation

- DROP the ODIN-tier filter from any options strategy. Use Gungnir probability for equity sizing but not options selection.
- If tier must be used, prefer T3 (n=13 is too thin to trust alone) + T4 — a REVERSAL of the standard filter.
- Better alternative: use ORATS term-structure tilt ($IV_{event_expiry} / IV_{far}$) as the primary filter. Tilt < 1.10 = flat curve = cheap event premium. Live-test this filter before going retrospective.

5. Key Strategy Evaluations

Three named strategies evaluated on the honest DTE-matched subset, with bootstrap CIs and top-k robustness.

Strategy: PDUFA Small/Mid Approvals (the announced playbook edge)

Fill model	n=92	Avg return	Win %	95% CI
MID (50% spread)	92	−2.30%	40.2%	[−14.24%, +10.10%]
REAL_40 (realistic)	92	−8.09%	38.0%	[−19.48%, +3.52%]

Median is −14.72% at MID. The strategy loses money at realistic execution and the CI at REAL_40 is [−19.48%, +3.52%], grazing but not crossing zero. Announced +6.4% was an artifact of the DTE-mismatch bug + optimistic fill assumption.

Verdict: PULL. Not a tradeable edge. This was the strategy featured on the playbook homepage.

Strategy: Phase 1/2 Positive Readout (the only robust edge)

Variant	n	Avg return	Win %	95% CI
MID (50%)	36	+45.1%	58.3%	[+12.0%, +80.5%]
REAL_40 (realistic)	36	+33.7%	—	—
MID, ex top 5 trades	31	+16.4%	—	—

This is the cleanest edge in the entire v1.2 dataset. CI lower bound +12% at MID. Survives top-5 removal. Survives 40% spread capture. The mechanism is clean: Phase 1/2 biotechs are not yet on institutional radars, options are cheap, positive data is genuinely surprising.

Verdict: KEEP and promote. Size modestly (n=36 means single-year risk is real). Diversify across ≥6 events per year. Cardinal rule: enter T-14, exit T-1.

Strategy: PDUFA Micro (all outcomes)

Variant	n	Avg return	Win %	95% CI
MID (full sample)	65	+34.07%	49.2%	[+6.11%, +66.88%]
REAL_40 (full sample)	65	+24.95%	46.2%	[−1.64%, +57.18%]
MID, ex top 2 trades	63	+18.16%	47.6%	—
MID, ex top 5 trades	60	+4.27%	45.0%	—

Median is 0.0% at MID and −7.97% at REAL_40. The +34% average is almost entirely driven by ~5 moonshots. Strip 5 trades and you're at +4.27% with 45% win. This is a lottery-ticket strategy, not an edge.

Verdict: OPTIONAL. Only trade if you can tolerate 55% losers per moonshot and very small per-position sizing (0.5–1% max). The REAL_40 CI lower bound of –1.64% means even at realistic execution the expected value isn't cleanly positive.

6. What This Changes for Q2 2026 Strategy

The current portfolio is GRCE (PDUFA Apr 23), WHWK (AACR), CRDF (AACR), CABA (AAN + EULAR), ALXO (ESMO May 7). The v1.2 results reshape how options should overlay that portfolio.

Do trade options on

- Phase 1/2 positive readouts that meet our Gungnir probability threshold (verified robust, +45% avg, +16% ex-top-5).
- PDUFA micro-cap catalysts — but ONLY with 0.5–1% position sizing, and ONLY if you can stomach the moonshot-or-nothing distribution. Do not scale this beyond a lottery-ticket bucket.

Do NOT trade options on

- PDUFA Small/Mid Approvals as a blanket strategy — the announced edge is not real after DTE fix.
- ANY Phase 2b or Phase 3 readout — Phase 3 ALL at -11.1% MID, Phase 2b at -31.7%. Stock may move but options consistently lose.
- PDUFA Large/CRL — worst sub-segment in the dataset at -64.2% MID.
- Nano-cap catalysts via options — they're not in the dataset at all. Spreads in nano cap chains will be even wider than micro.

Filter changes

- DROP the ODIN T1/T2 filter for options. It actively destroys edge (21 pp gap vs T3/T4).
- REPLACE with ORATS term-structure tilt filter: prefer tilt ≤ 1.10 (flat / cheap event vol). Untested retrospectively but theoretically sound.
- ADD bootstrap CI reporting to every published playbook number.
- ADD MID + REAL_40 side-by-side to every published playbook number.

Current portfolio implications

- GRCE (PDUFA Apr 23, micro-cap): options overlay is consistent with the PDUFA Micro edge. UOA 10/10 SCREAMING BULLISH has positional support. But the edge is tail-driven; size 1% max on the options leg.
- WHWK / CRDF (AACR posters): these are conference catalysts, not Phase 1/2 readouts per se. The conference overlay signal (+20% AACR boost) applies to equity, not directly to options. No options overlay recommended beyond current equity positions.
- CABA (AAN oral + EULAR): the oral presentation in autoimmune is closer to a Phase 2 readout. Phase 2 at MID is +2.1%, not a clean edge. Recommend NO options overlay; hold equity per existing thesis.
- ALXO (ESMO Breast May 7): this is an oncology Phase 2 readout. Same logic — Phase 2 overall is coin-flip. Recommend no options overlay. Smart Money flag (venBio \$5M) supports equity not options.

7. Methodology Appendix

Filter chain

- Raw: 1,828 trades (795 PDUFA + 1,033 Readout) from options_backtest_v2_results.json
- has_valid_prices: require entry_bid/ask > 0 and exit_bid/ask > 0. Kept 1,828.
- iv_sanitized: drop rows where |entry_iv_pct| or |exit_iv_pct| $\notin$ (0.01, 500) or |iv_change_pct| > 1000. Kept 1,768 (-60).
- dte_matched: entry_dte - exit_dte $\in$ [13, 23] AND exit_dte $\geq$ 3. Kept 804 (-964).
- Final honest subset: 804 trades. 322 PDUFA, 482 Readout.

Fill models

- WORST (0% capture): entry at ask, exit at bid. This is the naive urgency fill.
- REAL_25 (25% capture): entry at bid + 0.75 $\times$ (ask-bid), exit at bid + 0.25 $\times$ (ask-bid). Conservative paced limits.
- REAL_40 (40% capture): standard realistic paced-limit assumption for small-cap biotech options.
- MID (50% capture): entry and exit at mid price. Textbook but often unachievable in illiquid biotech chains.

Bootstrap

- n_boot=2000, seed=42 for reproducibility.
- Percentile method: 2.5th and 97.5th percentiles of the bootstrap mean distribution.
- CI-crosses-zero flag: CI lower bound $\leq$ 0 AND CI upper bound $\geq$ 0.

Files shipped with this memo

- options_backtest_v12_rerun.py — recompute pipeline (~340 lines)
- options_backtest_v12_results.json — segment-level JSON with every fill model and CI
- build_options_v12_docx.js — this memo's generator
- Q2_2026_Options_v1_2_Honest_Results.docx — this memo

What v1.2 did NOT re-test

- Survivorship bias (spread>55% exclusion): still present. A v1.3 shadow backtest on the >55%-spread universe is the next step.
- Nano-cap universe: still excluded. Needs separate ORATS pull.
- IV expansion curve: needs rerun with the parser fix. Deferred.
- ORATS term-structure tilt as primary filter: not backtested here because ORATS historical term-structure cache isn't built. Priority for v1.3.
- FinBrain PCR + insider flow as features: blocked by FinBrain MCP Pydantic bug on req parameter.

— End of v1.2 Honest Results Memo —

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